

District Crime Risk Tiering

An ETL + inference pipeline over NCRB crime data (data.gov.in)

1. Overview

This project builds an end-to-end ETL pipeline that ingests district- and state-wise crime data published by the National Crime Records Bureau (NCRB) on data.gov.in, normalizes it into a clean, queryable warehouse, and adds an inference layer that classifies each district into a risk tier (Low / Medium / High) per crime category and year. The goal is to demonstrate a realistic ETL workflow -- multi-source ingestion, schema normalization, feature engineering, and a lightweight ML/statistical inference stage -- rather than a simple load-and-plot exercise.

2. Data Sources (data.gov.in)

- District-wise IPC crimes (Indian Penal Code categories)
- Crimes against women (separate catalog / schema)
- Crimes against children (separate catalog / schema)
- Population / Census estimates, for per-capita normalization

3. Pipeline Architecture

1. Extract -> pull NCRB catalogs via data.gov.in resource API (JSON/CSV)
2. Transform -> reshape wide -> tidy long format
normalize district/state names (handle boundary changes)
normalize crime-type labels across years
join population data -> compute rate per 100k
3. Feature -> YoY % change, 3-year rolling average, in-state rank
4. Classify -> tier assignment (rule-based quantiles, then k-means)
5. Load -> warehouse tables (star schema, see below)

4. Warehouse Schema

```
dim_district(district_id, district_name, state_id)
dim_crime_type(crime_type_id, category, subcategory)
fact_crime(district_id, year, crime_type_id, count,
           population, rate_per_100k)
fact_crime_risk(district_id, year, crime_type_id,
                method, tier, score)
```

5. Risk Classification Approach

Two methods are computed side by side (distinguished by the 'method' column), so results can be compared:

- Rule-based (quantile): within each state + year + crime category, districts are split into tertiles by rate-per-100k -> bottom third Low, middle Medium, top third High. Simple, transparent, used as a baseline.
- Model-based (k-means / GMM, k=3): clusters districts per crime category using features (rate_per_100k, YoY change, 3yr rolling average). Cluster centroids are sorted by mean rate post-hoc and mapped to Low/Medium/High, since raw cluster IDs are unordered. Distance-to-centroid is stored

as a rough confidence score.

6. Key Data-Quality Challenges

- District boundary changes over time -- new districts are carved out between reporting years, breaking naive year-over-year comparisons; requires a crosswalk table or exclusion of affected districts from trend features.
- Crime category drift -- NCRB has renamed/merged reporting categories across years, requiring an explicit category-mapping table to normalize labels.
- Small-population volatility -- low-population districts can show large swings in per-capita rate from a single incident; a minimum population threshold or smoothing is applied before tiering.
- State-relative vs. national tiering -- tiers are computed relative to each state's own distribution (not a single national scale), since baseline crime rates vary structurally by state.

7. Suggested Output / Deliverable

A queryable table (fact_crime_risk) enabling questions such as: which districts moved from Medium to High risk for crimes against women between two reporting years; which districts are consistently Low risk across all categories; and how rule-based and model-based tiers agree or diverge for a given district-year.